

# Chaitali Ranalkar

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## RESEARCH INTERESTS

- Data systems for AI: LLM-assisted cleaning, querying and feature engineering over messy tabular data
- Machine learning on tabular data: tree ensembles (random forest, gradient boosting, XGBoost) and clustering (K-means, GMM, DBSCAN), and understanding when each model's assumptions actually hold
- Deep learning for time series: sequence models such as LSTMs versus statistical baselines (ARIMA, Prophet) on non-stationary data
- Reliable, interpretable models: uncertainty estimation, probability calibration and SHAP explanations

## EDUCATION

**St. Xavier's College (Autonomous), Mumbai**

M.Sc. in Big Data Analytics

**Aug 2023 – Jun 2025**

CGPA: 9.35 / 10

**B. N. Bandedkar College**

B.Sc. in Mathematics

**Jun 2020 – Jun 2023**

CGPA: 9.25 / 10

## RELEVANT PROJECTS

**Quick-Commerce Operations Gap Analysis** — *Independent Project · Python, BigQuery, Power BI*

**Sep 2026**

- Modelled a simulated 10-minute grocery-delivery platform (dark stores, inventory, delivery times, refunds, support tickets) to find where customer demand and revenue leak across the order lifecycle.
- Found that 54.3% of 150,000 searches ended with the product unavailable, so the largest loss occurs *before* an order is placed rather than during delivery; built Power BI views for platform health, leakage by area, and refund and delay follow-up.
- Traced an outlier store (6.1% on-time vs. a 42% network average) through a documented 26-query BigQuery root-cause chain, ruling out four hypotheses (peak-hour load, distance, workload, picking) and flagging it for on-site review instead of guessing a cause.

**Customer Segmentation with RFM Features and K-Means** — *M.Sc. Research Project, Dept. of IT*

**Oct 2024**

- Engineered log-transformed, min-max normalized Recency-Frequency-Monetary features from raw transactions; K-means (elbow at  $K = 3$ ) produced three interpretable segments, benchmarked against rule-based RFM quintile scoring.
- Showed the two methods agree at the extremes but diverge in the middle, and analyzed where K-means' assumptions break (non-spherical clusters, outlier-driven centroids), motivating GMM, DBSCAN and hierarchical alternatives.

**Predicting Customer Churn: Five Classifier Families Compared** — *M.Sc. Coursework*

**Mar 2024**

- On 10,000 bank customers, a random forest tuned with 5-fold cross-validated search reached 88.0% accuracy, F1 0.83 and ROC-AUC 0.94, against 75.7% for logistic regression, which points to a non-linear decision boundary.
- Identified age as the dominant predictor (importance  $\approx 0.28$ , about twice the next feature), then examined the limits of that finding: impurity-importance bias, class imbalance ( $\sim 20\%$  churners) and uncalibrated probabilities.

Forecasting ETH-USD: LSTM vs. Statistical Baselines — TensorFlow/Keras, statsmodels, Prophet

Jan 2024

- Modelled 1,827 daily closes of a non-stationary series (ADF  $p = 0.34$ ); the LSTM reached a test RMSE of 135.8 vs. 1,031 (SARIMA), 1,040 (Holt-Winters) and 1,594 (Prophet), capturing turning points rather than only the trend.

PROFESSIONAL EXPERIENCE

Senior Executive, KKC & Associates LLP

Jul 2025 – Present | Pune

- Built an ITGC automation tool (Python, Streamlit) with a fully on-premises NLP pipeline for document summarization and policy search, so confidential audit data never leaves the firm; manual review effort fell by 50%.
- Developed Python tools that extract and consolidate data from many unstructured PDFs into clean, structured datasets for analysis.
- Automated ITGC testing and GST R1/3B reconciliation (extraction, validation, mismatch reporting) for 100+ reconciliations per cycle, cutting manual effort by 40%.
- Designed a Power BI/DAX project and partner management dashboard with row-level security, year-over-year cost comparison, and ITGC/ITAC/DA breakdowns of billing and utilization.

Data Analyst Intern, KKC & Associates LLP

Jan 2025 – Jun 2025 | Mumbai

- Analyzed audit data with SQL, Power BI, Excel and CaseWare IDEA; built analytics covering financial performance, partner utilization, billing and project tracking.
- Prototyped the Python NLP summarizer for long policy and audit documents that was later developed into the ITGC tool.

POSTER PRESENTATION

Applications of Fourier Series — B.Sc. Mathematics · Third Prize

2020 – 2023

TECHNICAL SKILLS

|                               |                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Programming                   | Python, R, SQL, DAX                                                                                                       |
| Machine Learning & Statistics | scikit-learn, XGBoost, TensorFlow/Keras, statsmodels, Prophet; clustering, ensemble methods, time-series forecasting, NLP |
| Generative AI                 | Prompt engineering, LLM-assisted SQL generation                                                                           |
| Data & Cloud Platforms        | Google BigQuery, Google Cloud, Google Colab                                                                               |
| Analytics & Tools             | Power BI, Tableau, Streamlit, Excel, CaseWare IDEA, Power Automate                                                        |

CERTIFICATIONS

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|--------------------------------------------------------------------------|----------|
| Generative AI SQL Database Specialist — Vanderbilt University (Coursera) | Jul 2026 |
| Introduction to Cloud Computing — Simplilearn                            | Jan 2024 |
| Machine Learning — Great Learning Academy                                | Dec 2023 |